

Received 24 April 2026, accepted 7 May 2026, date of publication 15 May 2026, date of current version 28 May 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3693893

RESEARCH ARTICLE

A Generic Explainable Approach Model for Energy Forecasting in Smart Buildings

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ABSTRACT As the demand for energy-efficient buildings increases, accurately predicting and controlling energy consumption in smart buildings has become a critical challenge. studies have addressed energy consumption forecasting using AI models; however, these approaches often lack transparency and interpretability. To address this challenge, eXplainable Artificial Intelligence provides the tools necessary to enhance the interpretability of these predictive models. This study integrates machine learning and Explainable Artificial Intelligence (XAI) techniques to develop an explainable approach to analyze the impact of energy-related factors on energy consumption within smart building environments. We utilize three datasets Lighting, HVAC and PPLs, collected from a smart building in Virginia, USA. A wide range of ML models are employed to predict electricity usage, while three different XAI techniques (SHapley Additive exPlanations, Local-Interpretable Model-Agnostic Explanations(LIME) techniques and Explain Like I'm 5 (ELi5)) are used to interpret the model outputs from both global and local perspectives. The findings reveal that the proposed framework enhances model interpretability and supports the selection of high-performing predictive models. The explanations provided by explainable AI techniques, including SHAP, LIME, and ELI5, at both local and global levels, were consistent with the results observed during the exploratory data analysis. Although this study focuses on energy consumption data, the proposed methodology may be extended to other data-driven smart city applications, such as intelligent transportation systems.

INDEX TERMS Electricity consumption, ELI5, energy, explainable AI, LIME, SHAP, smart buildings.

I. INTRODUCTION

By 2050, residents in urban areas will constitute over 68% of the global population [1], resulting in high demand for transportation, water, and energy use. This rapid urbanization requires a significant transformation in the way cities are designed and managed to provide the sustainability and well-being of all citizens in those areas [4]. As the urban population expands, policymakers and managers face significant challenges in addressing this growth to achieve a long-term success. For this, effective energy resource management becomes a must by deploying efficient policies for public transportation to reduce both traffic congestion [5]

The associate editor coordinating the review of this manuscript and approving it for publication was Zhengmao Li^{1b}.

and carbon emissions [6]. Furthermore, decision-makers need to develop strategies to manage water and energy resources in a way that does not exacerbate environmental degradation. To achieve the aforementioned goals, a sustainable urban area must be developed to meet the demands of the growing populations while maintaining resilient and adaptable for future challenges. To address these issues, the transition toward smart cities becomes essential [7]. The integration of advanced technologies, including artificial intelligence (AI) systems and the Internet of Things (IoT) [10] forms the pillar for transforming conventional cities into intelligent, sustainable, and livable environments which are capable of managing resources, enduring natural disasters, and other catastrophic phenomena [2]. Smart cities increasingly require the deployment of AI-powered systems

across diverse domains, including transportation, road traffic management [8], smart grids [9], and innovative waste management [11]. Consequently, a vast amount of real-time data collected using IoT sensors must be processed to extract insights and build predictive models to optimize resource consumption for a sustainable environment.

Despite significant advances in AI algorithms, traditional ML models often operate as non-interpretable models, failing to provide transparency on the overall process behind their decisions. Their lack of interpretability significantly leads stakeholders to be unconvinced of adopting these systems. To address this problem, researchers explored another variant of AI, called explainable AI (XAI) [3], to make AI models more understandable and interpretable, thereby enhancing users' trust and confidence. XAI refers to techniques designed to explain the decision-making process underlying model predictions [17], [18], thus promoting a human-centered approach that places the end-user at the core of AI systems [15].

Deep learning models are widely used in smart city applications due to their ability to model complex non-linear relationships. These models are considered opaque due to their lack of interpretability [21]. The integration of XAI into these models becomes essential in highly sensitive domains, including healthcare, finance, and transportation, where predictions must be both accurate and explainable.

The energy sector is a broad field in which advanced AI algorithms are used to optimize energy consumption [14], and this requires deploying predictive models capable of striking a balance between accuracy and interpretability. To foster user confidence in these systems, it is essential to provide explanations of the predictions by identifying the discriminative features using XAI techniques [13].

In our paper, we present a novel interpretable approach to forecast energy consumption in smart buildings. By integrating XAI techniques and ML models, we were able to deliver both performant and interpretable models. XAI techniques play a pivotal role in identifying influential features at both global and local levels using SHapley Additive Explanations (SHAP), Local interpretable model agnostic explanations (LIME) and Explain like I am 5 (Eli5) techniques. The aforementioned techniques elucidate the factors (HVAC, PPLs, and Lighting) that affect energy consumption. This approach allows end users, including stakeholders, data scientists, energy experts, and policymakers, to gain deeper insights into the resulting output and take further measurements in the future. Ultimately, the explainable approach bridges the gap between complex data-driven algorithms and practical, comprehensible applications in energy management domains.

To guide this study, our main hypothesis is that explainable machine learning models can achieve competitive predictive performance while providing interpretability and supporting informed decision-making. This main hypothesis is examined through the following sub-hypotheses:

- **H1:** Explainable machine learning models can achieve competitive predictive performance compared to traditional opaque approaches.
- **H2:** Explainable AI techniques can reliably identify the most influential energy-consuming components, such as HVAC systems, lighting, and plug loads.
- **H3:** The use of explainable AI enhances transparency and trust, making energy forecasting models more suitable for stakeholder-oriented decision support.

This paper is structured as follows:

- The second part explores the shift from traditional AI techniques to Explainable AI, emphasizing the growing demand for transparency and interpretability in AI models. Then, highlighting how XAI is being applied across various sectors, demonstrating its impact on several Sustainable Development Goals (SDGs).
- The third part of the paper presents a case study that identifies the role of building our AI models on an explainable approach to foster user trust. By predicting electricity consumption in smart buildings and identifying the features that consume more electricity.
- Finally, we will discuss the results of this case study, evaluating the benefits of XAI in this specific context and its potential for broader applications.

LIST OF ACRONYMS

AI	Artificial Intelligence.
XAI	Explainable Artificial Intelligence.
ML	Machine Learning.
DL	Deep Learning.
MLP	Multi-Layer Perceptron.
DTR	Decision Tree Regressor.
SVR	Support Vector Regressor.
RF	Random Forest.
GBR	Gradient Boosting Regressor.
XGBoost	Extreme Gradient Boosting.
LIME	Local Interpretable Model-Agnostic Explanations.
SHAP	SHapley Additive exPlanations.
ELI5	Explain Like I'm 5 (XAI toolkit).
MAE	Mean Absolute Error.
RMSE	Root Mean Squared Error.
R^2	Coefficient of Determination.
EDA	Exploratory Data Analysis.
NMBE	Normalized Mean Bias Error.
CVRMSE	Coefficient of Variation of the Root Mean Squared Error.
SDG	Sustainable Development Goals.
IoT	Internet of Things.

II. BACKGROUND AND RELATED WORKS

A. FROM TRADITIONAL AI TO EXPLAINABLE AI (XAI)

Over the past decade, AI and ML algorithms have been used across various domains to create predictive systems. Although these models often achieve strong performance,

they function as opaque due to their lack of transparency, making it difficult for end users to understand how they produce outputs. To address this limitation, a new field of AI has emerged with explainability as its fundamental focus: known as Explainable Artificial Intelligence (XAI). XAI bridges the gap between predictive performance and transparency [22].

To understand the concept of XAI, it is essential to clarify the difference between explainability and interpretability. Explainability is the process of generating explanations to understand a model's output in a human-understandable manner [23]. While interpretability, refers to the way in which the rationale behind predictions is identified by examining the relationships between inputs (features) and outputs (predictions). In predictive modeling, explainability has two complementary levels: local and global. Local explanations rely on explicating individual predictions. While global explanations focus on the model's overall functioning across all inputs, revealing insights into its general behavior.

Furthermore, the use of explainable methods relies on the underlying model. Some techniques are model-specific and designed to explain particular types of models. Contrarily, model-agnostic methods can be used across a wide range of models to generate explanations. Both approaches contribute to improved model transparency and accountability.

B. XAI IN SMART CITIES AND SUSTAINABILITY

1) XAI FOR SMART CITY APPLICATIONS

A smart city is a novel concept that involves the strategic application of futuristic infrastructure (internet of things and big data analytics) to maximize urban utilities in various aspects and enhance the quality of life. AI in public administration supports this vision by enabling data-driven decision-making, automating routine services, and personalizing citizen interactions, thereby improving efficiency, transparency, and public value creation [16]. However, its diffusion also raises ethical tensions such as automation versus augmentation and data privacy versus security, which must be managed through robust governance frameworks [16]. These technologies are used to capture, in real time, data from various sensors and devices throughout the city, and to act on that information. From a European perspective, smart cities are defined as an integrated urban system in which advanced digital technologies, coordinated governance strategies, and supportive legal frameworks are jointly leveraged to improve the efficiency, sustainability and resilience of public services [12]. Artificial intelligence takes center stage, using machine learning and deep learning tools to train its algorithms, enabling the development of predictive models that help decision-makers improve urban operations. AI can leverage state-of-the-art technologies in smart city applications to reduce traffic congestion and improve energy efficiency in renewable energy systems [24]. AI models used in smart city applications operate as opaque models

due to their intricate decision-making processes, so end-users cannot rely on the generated predictions if they do not understand the processes behind [25]. To enhance users' trust and reliability in those systems, an innovative branch of AI, eXplainable Artificial Intelligence (XAI), has been developed to explain the rationale behind these processes and make them more understandable to city officials and the public.

2) XAI AND THE SUSTAINABLE DEVELOPMENT GOALS (SDGs)

The United Nations instituted the 17 Sustainable Development Goals (SDGs) in 2015 to counter such rising global socio-economic and environmental challenges, with the goal of creating a fairer and more sustainable approach for future generations. It is estimated that the realization of Sustainable Development Goals (SDGs) is linked with smart city solutions like Smart Transportation, Smart Buildings, Public Safety, and Intrusion Detection. These are solutions that look to increase urban efficiency, improve the quality of life for city residents, and help to create more environmentally-friendly cities – all key focuses within the Sustainable Development Goals created by the United Nations.

3) XAI FOR INDUSTRY, INNOVATION AND INFRASTRUCTURE (SDG 9 AND SDG 11)

Smart transportation and autonomous vehicles:

One of the top emerging and rapidly evolving trends in the automotive field is the development of autonomous driving, which sets the tone for the industry's future and makes it a prominent headline. The deployment of advanced, autonomous technological vehicles requires a highly meticulous and careful approach, because any lack of precision or mistake could lead to the potential loss of human lives and fatal consequences [26]. Explainable AI has had a transformative impact on smart transportation by providing comprehensive explanations of decision-making processes, fostering trust and accountability in these systems, including traffic pattern prediction and optimal route suggestions. These technologies have contributed to reduced congestion and improved overall urban mobility. Researchers in the study [27] highlight the use of a transformer-based model to generate textual descriptions of images through an explainable multi-modal deep learning framework. This model provides textual explanations of the predicted actions by exploiting both transformer architectures to generate descriptive text and attention mechanisms to highlight the salient regions that influence the model's predictions for a given image.

Another study conducted by [28] highlights the development of a model that leverages Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) as the encoder and decoder components, respectively, to enhance the model's robustness and thereby increase user

TABLE 1. Overview of recent studies applying XAI in smart city and energy-related applications.

Study	Application	Methodology	ML Models	XAI Techniques	Use of XAI
[28]	AV	Attention-based approach highlighting regions influencing predictions	LSTM (decoder), CNN (encoder)	Visual attention	Improve model performance
[27]	AV	Transformer-based image-text explanation model	DNN	Attention-based model	Enhance user trust, improve safety, ensure accountability
[29]	Smart transportation	Predict public transport usage; identify important features	Tree-based models	SHAP	Identify factors most influencing user demand
[30]	Road traffic management	Identify the effect of road transport on groundwater chemistry	XGB	XAI (feature importance, ALE)	Provide insight into impact of road transport variables
[31]	Flood risk management	Predict flood susceptibility and explain influencing factors	CNN	SHAP	Feature contribution
[32]	Flood risk management	Identify feature impact before and after dam construction	LSTM	SHAP	Identify factors affecting flood phenomena
[33]	Energy consumption	Select optimal features	XGB, LightGBM, LSTM, SVR	SHAP	Improve model performance
[34]	Forecast energy use	Identify feature influence	XGB	SHAP	Feature contribution
[35]	Total energy consumption	Identify feature importance	XCNN, LSTM	SHAP	Reveal feature influence on predictions
[37]	Forecast energy consumption	Detect relevant features	LSBoost, SVR	SHAP	Improve model quality
[38]	PV power forecasting	Exclude irrelevant features to improve model	Random forest regressor	SHAP, LIME, ELI5	Improve model quality
[40]	PV generation	Model optimization by removing low-impact features	XGB	ELI5	Improve model quality
[41]	Residential load clustering	Determine effect of each feature per class	Clustering methods	SHAP	Feature contribution
[42]	Offshore wind energy	Identify significant influence features	RF	SHAP, PDP	Feature contribution
[43]	Substation load forecasting	Contribution of each feature	Semi-supervised algorithm	SHAP	Feature contribution
[44]	Snow load outage estimation	Contribution of features	RF	SHAP	Feature contribution
[45]	Building design energy efficiency	Contribution of each feature	NGBoost	SHAP	Feature contribution
[46]	Dew point cooling performance	Contribution of each feature	DNN	SHAP	Feature contribution
[48]	Cooling energy consumption	Contribution of each feature to illustrate climate impact	—	SHAP	Feature contribution
[49]	CO ₂ emissions forecasting	Identify factors influencing consumption and emissions	RDF, SVM, AdaBoost, ANN, KNN, GBDT	SHAP (feature selection)	Feature contribution
[50]	Electric bus energy prediction	Identify key factors	LSTM-XGB	SHAP	Feature contribution
[51]	Stirred tank reactor energy	Identify impact of geometrical parameters	ANN	LIME	Feature contribution
[52]	Solid yield prediction	Identify most influential features	GTB, RF, SVM	SHAP	Feature contribution
[53]	Heat demand forecasting	Interpret opaque model	ANN	SHAP	Feature contribution
[39]	Combustion characteristics	Extract most relevant features	RF, AdaBoost, ANN	SHAP	Feature contribution
[54]	Biogas-powered engines	Assess feature impact	ExtraTree, XGB, LightGBM	SHAP	Feature contribution

trust. Moreover, some researchers in study [34] explored the integration of XAI into smart transportation and road traffic management by underscoring the significant applications of XAI and ML models in predicting public transportation

usage rates. The researchers employed a tree-based model and SHAP to identify cross-feature interactions that drive the model's predictions, thereby improving transparency and interpretability of the transportation forecasting system.

A recent study by [47] develops a Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) models as encoder and decoder components to enhance the robustness of the framework and increase user trust. Moreover, some researchers in study [29] explored the integration of XAI in smart transportation and road traffic management by underscoring the significant application of XAI and ML models in predicting public transportation usage rates. The researchers employed a tree-based model and SHAP to identify cross-feature interactions that drive the model's predictions, thereby improving the transparency and interpretability of the transportation forecasting system. On road traffic management, [30] has investigated the impact of road transport on groundwater chemistry parameters by integrating XAI techniques (feature importance and ALE) to elucidate the influence of transportation on groundwater parameters and employing multivariate analysis to identify the chemistry of the selected groundwater parameters using Extreme Gradient Boosting (XGB). These advancements not only streamline transportation but also contribute to sustainable urban development by reducing emissions and energy consumption. These advancements not only improve transportation management but also contribute to sustainable urban development by reducing emissions and energy consumption.

4) XAI FOR WATER, CLIMATE, AND MARINE ECOSYSTEMS (SDG 6, SDG 13, SDG 14)

In several studies, AI models have been used in environmental monitoring to analyze satellite imagery to track marine regions affected by pollution. This approach is widely adopted to understand the areas of the sea that contain much waste and how mitigation efforts can be applied. The use of Explainable artificial intelligence (XAI) and machine learning algorithms can enhance user trust and extract latent spatio-temporal features relevant to geospatial analysis. In that case, resource and safety management is an emerging research area to face this problem. The use of satellite imagery to identify ocean and coastal regions affected by pollution in near real time, supports more targeted mitigation strategies. Similar approaches have been applied in flood monitoring, where XAI and ML help identify factors contributing to increased flood risk.

The authors in [31] employ explainable methods, specifically SHAP and a CNN model, to forecast flood susceptibility. CNN is used to analyze spatial data. SHAP, on the other hand, shows how different flood risk factors affect outcomes. Another study by [32] demonstrated that the XSHAP technique can offer insights into the primary factors contributing to flooding before and after construction of the Three Gorges Dam. This approach may also provide indications, though not definitive ones, about the dam's effectiveness in reducing downstream flooding. The integration of XAI techniques across diverse domains, including transportation and environmental monitoring, enhances user

trust. In some cases, it may also improve model quality. Beyond performance considerations, these techniques can create space for scientific exploration when data structures are complex.

C. OVERVIEW OF RESEARCH STUDIES IN THE ENERGY SECTOR

Integrating XAI into the energy landscape is essential, as the energy sector represents a critical domain that directly affects human well-being. By identifying the key factors driving energy overuse, it becomes possible to support more sustainable energy management practices and enhance transparency in decision-making processes. This approach helps optimize energy resource utilization and reduce overall consumption. In addition, building energy consumption is strongly influenced by climatic conditions, particularly through their impact on heating, ventilation, and air conditioning (HVAC) systems. External environmental variables such as outdoor temperature, humidity, and solar radiation significantly affect heating and cooling loads in commercial buildings, often making HVAC systems the dominant contributor to total electricity demand [34], [48]. Situating explainable machine learning analyses within this context is therefore important for ensuring that model interpretations remain consistent with the known physical and operational behavior of building energy systems.

In a study conducted by [33], the authors developed an interpretable model for energy consumption forecasting using machine learning techniques including XGBoost, LightGBM, LSTM, and Support Vector Regressor, combined with explainable AI (XAI) methods. These XAI techniques were employed to identify the key features influencing the model's predictions, enabling the selection of the most relevant variables for the forecasting task. Similarly, [34] employed XGBoost and SHAP techniques to forecast energy consumption in both commercial and residential buildings, providing deeper insights into the factors affecting energy demand.

More recently, [35] proposed an explainable deep learning framework combining convolutional neural networks (CNNs) and long short-term memory (LSTM) networks to predict energy consumption. In their approach, an autoencoder was first used to reduce data dimensionality, after which CNN layers extracted significant features from the reconstructed data. LSTM networks were subsequently employed to capture temporal dependencies between the extracted features and the target variable.

Additional studies, such as [36] and [37], explored predictive modelling of energy consumption using machine learning approaches combined with explainable AI techniques. These works focused on identifying and eliminating irrelevant features to improve the overall accuracy of energy consumption prediction models. Several studies [28], [33], [37], [38], [39], [40] have further demonstrated that explainability methods can enhance both model performance

and robustness by identifying influential variables and interpreting their effects on predictions. Moreover, researchers have investigated feature contributions and interactions using XAI techniques across a variety of energy-related applications [41], [42], [43], [44], [45], [46], [48], [49], [50], [51], [52], [53], [54].

The works cited above, summarized in Table 1, primarily focus on applying one or more XAI techniques to interpret machine learning models. In contrast, the present study adopts a comprehensive approach by integrating multiple machine learning models and explainability techniques within a unified framework. While many previous studies rely on feature elimination during model optimization, our research retains all variables to preserve the complete information contained in the dataset. This inclusive strategy ensures that our interpretations account for all variables, potentially revealing information overlooked during feature elimination.

III. PREDICTIVE MODEL FOR ENERGY CONSUMPTION

This section examines the integration of ML models and XAI techniques to forecast energy consumption. Previous studies have leveraged various models, along with XAI techniques as discussed in section II. Our main study follows a structured methodological framework for developing an explanatory-predictive model for electricity consumption. We used analytical methods to identify the relationships between input features and electricity demand by exploiting the EDA analysis. Then, a comparative phase is adopted to evaluate the predictive performance of multiple ML models and XAI techniques. At the final stage, we employed inductive reasoning to derive insights from observed data patterns and explainability results to identify dominant energy-consuming components, while deductive reasoning was used to validate the proposed research hypotheses.

The objective of the present study is to propose a novel approach to developing an explainable predictive framework for electricity consumption forecasting. Furthermore, XAI techniques are incorporated into the model construction process to ensure the model is understandable. This aspect is essential throughout the modeling process rather than considered only during post-hoc analysis. Through this approach, we offer a comprehensible path through all stages of our decision-making processes. This approach not only builds trust and transparency but also provides more actionable information, enabling stakeholders to better understand the factors that influence predictions and make informed decisions.

A. METHODOLOGIES

Our proposed model adopts a comprehensive approach to develop an interpretable and explainable model, encompassing the entire process from data acquisition to model deployment.

1) MACHINE LEARNING MODELS

Multi-Layer Perceptron (MLP) is a type of Artificial Neural Network inspired by the human brain's cells. It consists of a feedforward artificial neural network composed of an input layer, one or more hidden layers, and an output layer, with neurons connected by weighted links and trained using backpropagation.

Decision tree regressor is a supervised learning model for regression tasks that offers high interpretability. It works by recursively dividing the dataset into subsets based on feature values to maximize information gain or minimize prediction error. The structure of the resulting model is similar to a tree, with each internal node representing a decision rule for a feature, the branches reflecting the outcomes of that rule, and the leaf nodes providing the final continuous predictions. It is possible to transform a decision tree into a set of human-readable decision rules, thereby facilitating its interpretability [55]. Generally, the construction of the tree is based on a division criterion that minimizes the mean square error (MSE) for regression problems [56].

Support vector regression (SVR) is a supervised learning model that performs regression by constructing an optimal function based on a subset of training samples called support vectors [20]. This model is designed to handle both linear and non-linear relationships via kernel functions and is known for its high generalization ability, particularly in high-dimensional spaces [19].

Random forest The Random Forest regressor is an ensemble learning model that uses bagging and random feature selection to combine multiple trained decision trees. This approach effectively reduces variance and overfitting while improving robustness and predictive accuracy, especially for complex and non-linear data.

Gradient Boosting Regressor is an ensemble learning method that constructs a strong model by successively combining multiple weak learners, commonly decision trees [45]. Each learner builds upon the previous ones, achieving high accuracy and robust predictive performance, thus making gradient boosting widely used in regression tasks.

eXtreme Gradient Boosting takes the principles of Gradient Boosting Regressor and extends them by providing an optimized and scalable implementation of gradient boosting. While both rely on combining weak learners to improve prediction accuracy, XGBoost further enhances this with regularization and computational optimizations, thereby achieving greater generalization and efficiency. Consequently, XGBoost is widely adopted for high-performance regression and prediction tasks [57].

2) XAI TECHNIQUES

It's important to note that z' in the LIME method refers to the perturbed instance, whereas z' in the SHAP method refers to a binary vector indicating the presence or absence of a feature in the explanation.

LIME is a technique for interpreting machine learning models by explaining predictions for individual instances. The method approximates the local behavior of a complex model using a simplified linear model that is both interpretable and capable of mimicking the behavior observed in the original model within the specified region [58]. By generating an interpretable representation that corresponds to the original set of inputs, then fitting a surrogate model around these new inputs to produce predictions that are closely aligned with those generated by the opaque model. The interpretable model minimizes the loss function as follows:

$$L(f, g, \pi_x) = \sum_{z, z' \in Z} \pi_x(z) (f(z) - g(z'))^2 \quad (1)$$

where: f is the non-interpretable model to be explained, g is the surrogate model used to describe the original model, x is the instance to be explained, L is the loss function, z is the original sample and z' is the perturbed sample, $\pi_x(z)$ is the proximity measure that calculates the distance between the perturbed sample and the original one. The explanation generated by LIME is as follows:

$$\xi(x) = \arg \min_{g \in G} [L(f, g, \pi_x) + \Omega(g)] \quad (2)$$

where ξ is the explanation, Ω is the model complexity, G is the set of linear models used to explain the f model.

SHAP is a global interpretation technique used to interpret opaque models. It is based on the game-theoretically optimal Shapley values [59]. The features are the players in a coalition game, the game is the instance under explanation, and the payoff is the prediction. The contribution of each feature of the prediction is calculated by computing the shapley values. The mathematical formula simplifies as follows:

$$g(z') = \phi_0 + \sum_{j=0}^M \phi_j z'_j \quad (3)$$

where g is the explanation model, $z' \in \{0, 1\}^M$ is the coalition vector; 0, 1 indicates the absence and the presence of a feature in the explanation, M is the set of features, ϕ_0 is the baseline value of the prediction and ϕ_j is the shapley value of feature j .

ELIS is a Python library that offers global explainability. It's based on feature selection by using permutation importance. Assigning each feature in the dataset an importance score that measures its contribution to the model's accuracy highlights each feature's significance in the model's prediction process.

3) EVALUATION METRICS

To understand clearly the performance of the models, we adopt an evaluation analysis using three different metrics RMSE, MAE, and provide different measures of accuracy,

which are defined by:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - y_{pred}| \quad (4)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - y_{pred})^2} \quad (5)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - y_{pred})^2}{\sum_{i=1}^n (y_i - y_{mean})^2} \quad (6)$$

Mean Absolute Error indicates the average magnitude of the model's prediction errors. At the same time, Root Mean Squared Error is more sensitive to larger errors, providing insights into the impact of significant deviations and the model's extreme behaviors. The coefficient of Determination R^2 provides an indication of how well the model predicts the target variable. In addition, normalized error metrics—namely the Normalized Mean Bias Error (NMBE) and the Coefficient of Variation of the Root Mean Squared Error (CVRMSE)—were incorporated to align the evaluation with established building energy performance assessment practices, as recommended by ASHRAE Guideline 14 [61] and the International Performance Measurement and Verification Protocol (IPMVP).

$$NMBE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)}{(n-1)\bar{y}} \times 100 \quad (7)$$

$$CVRMSE = \frac{RMSE}{\bar{y}} \times 100 \quad (8)$$

where y_i represents the measured electricity consumption, $\hat{y}_i$ denotes the predicted value, $\bar{y}$ is the mean of the measured values, and n is the number of observations.

B. OUR APPROACH

The dataset for this study was obtained from a 328 m² all-electric, zero-energy commercial building with two floors located in Virginia, USA.¹ Before analysis, the data were carefully prepared. The initial phase involved merging three separate data sets (lighting, PPL, and HVAC) into a single unified dataset. The forecasting problem is stated at the whole building level, with the target variable representing total hourly electricity use summed over the principal end-use categories, such as HVAC, lighting, and plug and process loads. The system boundary is specified as a partially closed system, where the internal electrical subsystems are represented in detail using consumption data, but external variables such as weather and occupancy are not modeled as input variables. Next, an explainable approach was adopted, starting with an understanding of the data set, since the interpretable model's predictions are fundamentally based on predictive characteristics. Using data visualization, we provide a comprehensive analysis of the HVAC, PPL, and lighting components that influence the building's electricity consumption. All features were retained during preprocessing

¹<https://osf.io/3kdhq/>

to maintain the integrity of the dataset and prevent bias that could result from their removal. Six machine learning models were deployed. The models employed were XGB, SVR, RF, MLP, DT, and GBR. Splitting the dataset into 17459 samples for training and 4365 samples for testing. Forecasting is formulated as a prediction for a given time, whose objective is to predict the hourly electricity consumption of the entire building at time $t+1$ based on historical observations up to time t . In order to avoid data leakage and maintain the temporal structure of the data, a TimeSeriesSplit cross-validation strategy was used, ensuring that the training data consistently precedes the validation data in time. Hyperparameter optimization was performed using GridSearchCV, with the coefficient of determination (R^2) employed as the primary optimization indicator, while the mean absolute error and mean squared error were used for additional performance evaluation. In addition to interpreting the overall prediction process, we explain model predictions using three techniques: SHAP, LIME, and feature importance results. The Eli5 library was employed to provide insights into the data and the models' performance. The XAI techniques were applied to the best-performing models after hyperparameter optimization. SHAP provided global and local feature importance interpretations; LIME provided case-specific explanations; and Eli5, with permutation importance, visualized feature weights and contributions to show the impact of shuffling features on the model's performance. By comparing the information obtained from these XAI techniques, we identified consistent models and unique contributors, thereby ensuring a comprehensive and reliable interpretation of the models' predictive performance. This multidimensional approach, which incorporates an explanatory methodology at each stage, provides a solid, detailed understanding of the data and model performance. The proposed explainable structure is model-independent and data-driven, based on generic machine learning models and a priori explainability techniques. Therefore, the structure is not related to a specific building configuration or geographical location and can be easily applied to other smart buildings or energy datasets with similar measurement structures. No building-specific adjustments are required, and the structure can be applied to different consumption profiles by simply adapting the input data. The proposed methodology is illustrated in Fig. 1.

1) DATASET

The dataset covers a 328 m² zero-energy, all-electric commercial building located in Virginia, USA, which is characterized by a temperate climate with seasonal heating and cooling demands. It consists of two floors. Information about the building includes electricity consumption, HVAC, lighting, electrical outlets, and PPLs. Although auxiliary weather information is available as part of the dataset, these variables were excluded from the study. This choice is aligned with our primary objective of focusing on explainability of measured internal electricity consumption. The data is

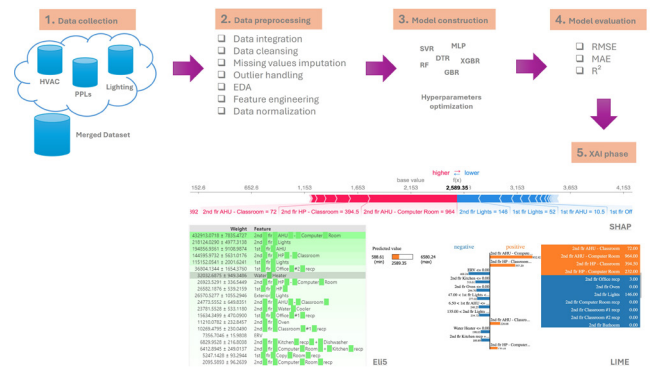


FIGURE 1. Proposed methodological framework for electricity consumption.

primarily composed of high-resolution energy measurements at a circuit level, whereas the weather data is offered as additional information. The climatic influences are thus implicitly reflected into the time series of electricity consumption, especially in the HVAC loads that show seasonal behavior. This enables the proposed explainable framework to identify and interpret the internal energy drivers without introducing new dependencies. The data was collected hourly between 18 June 2014 and 13 December 2016. The dataset comprises 21,824 samples and 36 features. The table below shows the datasets used and the HVAC, lighting, and PPL components for the two floors of the building. Table 2 provides information on the dataset's characteristics and components. This section, like the others, is based on the principle of interpretability; the characteristics presented are both significant for electricity consumption and easily understood by experts in the field. The dataset is publicly available via the Open Science Framework.²

2) DATA PREPROCESSING

The first step in the process is to combine the various data sources collected at different intervals across different characteristics into a single unified dataset. This was achieved by selecting the same time period for all data sources. Furthermore, exploratory data analysis (EDA) was integrated to facilitate examination through easily interpretable graphs. The techniques used at this step are as follows:

Data cleaning: First, we cleaned the data by identifying and removing duplicate records based on timestamps. Next, we imputed missing values, which can significantly impact the quality of the analysis and affect model performance. Outliers were then identified. These were retained to avoid losing potentially valuable information. The reason for this was that the outliers could be attributed to significant events, such as extreme weather, rather than simple measurement errors. To preserve data integrity, we considered it appropriate to include outliers arising from specific

²Project DOI: 10.17605/OSF.IO/3KDHQ.

TABLE 2. Description of dataset features.

Dataset	Features	
HVAC	Date	
	Total HVAC+HW energy use	
	Total HW energy use	
	Total HVAC energy use	
	HVAC-1 1st floor	1st flr AHU
		1st flr HP
	HVAC-2 2nd floor	2nd flr AHU – Classroom
		2nd flr AHU – Computer Room
		2nd flr HP – Classroom
		2nd flr HP – Computer Room
Lighting	Water Heater	
	ERV	
	Date	
	Total Lighting Energy use	
	Interior lights	
	Exterior lights	
PPLs	1st floor lights	
	2nd floor lights	
	Date	
	Total PPLs Energy use	
	1st floor PPLs	1st flr Lobby recp
		1st flr Office #1 recp
		1st flr Office #2 recp
		1st flr Office #3 recp
		1st flr Bathroom
		1st flr Kitchen
		1st flr Copy Room recp
		1st flr Utility Room recp
	2nd floor PPLs	2nd flr Office recp
		2nd flr Oven
		2nd flr Computer Room recp
		2nd flr Classroom #1 recp
		2nd flr Classroom #2 recp
		2nd flr Bathroom
		2nd flr Kitchen
		2nd flr Kitchen recp + Dishwasher
		2nd flr Water Cooler
		2nd flr Computer Room + Kitchen recp
		2nd flr Classroom #2 + Copy Room recp
		2nd flr Storage Room + Computer Room recp
		Refrigerator

environmental factors, particularly extreme weather conditions, that could affect the sustainability of the systems concerned.

Explanatory data analysis: Exploratory data analysis (EDA) is an essential technique in data preprocessing. This analysis provides a better understanding of the dataset and identifies relationships between the independent variables and the target variable. EDA processing relies on various visualizations, figures 2–4 illustrate the monthly distributions of HVAC and lighting systems, as well as their relative contribution to total electricity consumption over time.. Our analysis shows that the dataset does not follow a normal distribution. In addition, the distribution of each feature across months is shown to identify which features have a significant impact on electricity consumption. To analyse these seasonal effects, new features were engineered, including month, year, and season. The consumption pattern analysis based on the monthly and component-wise distribution of consumption reveals the typical consumption behavior of HVAC and lighting systems. The loads related to HVAC systems exhibit significant temporal variation, with distinct peaks during the heating and cooling seasons, thereby

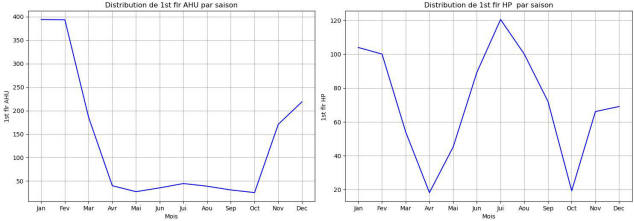


FIGURE 2. Distribution of HVAC (AHU, HP) for 1st floor.

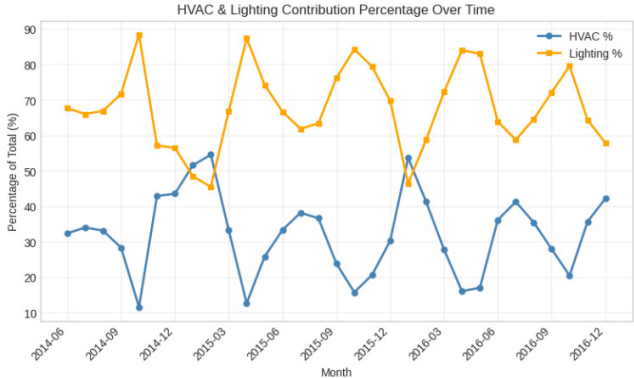


FIGURE 3. Distribution of HVAC (AHU, HP) and lighting per month.

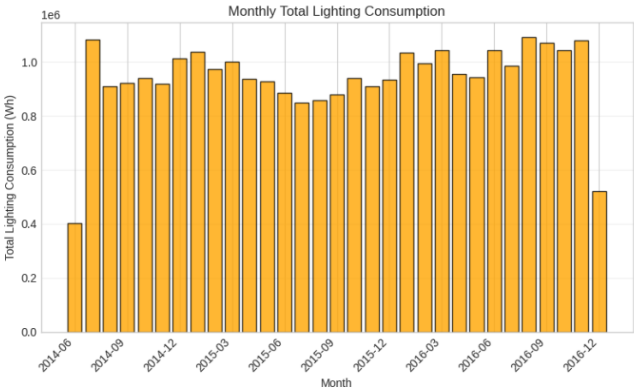


FIGURE 4. Distribution of lighting.

reflecting their significant dependence on the seasonal demands of their operation. Conversely, the lighting system’s consumption pattern remains relatively steady throughout the year, with moderate variation primarily established as a function of the occupancy and use schedules, and not as a function of climatic variations. In both floors, the HVAC system dominates the peak electricity demand, while the lighting system sustains the base load. These aspects are pivotal in comprehending the relative significance of the end-use components and establish the rationale for employing explainable AI techniques to quantify their influence on the prediction of electricity consumption. To better understand their impact on the model’s predictions, XAI techniques will be used in the following section to provide a detailed interpretation.

Data transformation: Two datasets were used to examine the role of data transformation on model performance. Normalisation was applied to the first dataset using Min-Max Scaler, bringing the features to the range [0,1]. In the second dataset, these features were normalised using StandardScaler. This approach aimed to verify the model's performance after normalisation.

IV. RESULTS AND DISCUSSIONS

This section adopts an inductive and comparative approach to present and examine the experimental results obtained through an explainable machine learning-based approach for predicting electricity consumption. Initially, a comparative analysis of predictions from several machine learning models was performed using standard evaluation metrics. Based on these results, a particularly high-performing model was selected for further analysis, enabling us to identify and interpret the characteristics that had the most significant impact on its predictions. Through this process, this study aims to explore how explainability techniques could contribute to both model performance and more transparent, interpretable decision-making.

A. COMPARATIVE MODEL PERFORMANCE

The optimal configuration of each machine learning model considered in this study was determined by optimizing the hyperparameters within the GridSearchCV framework. The search space for the parameters explored is summarized in Table 3. Through systematic adjustment, we identified high-performance configurations while ensuring a fair and consistent evaluation of all models.

We evaluated the model's performance using multiple regression metrics: root mean square error (RMSE), mean absolute error (MAE), coefficient of determination (R^2). Along with established regression model evaluation metrics, model performance is assessed using normalized mean bias error (NMBE) and coefficient of variation of the root mean squared error (CVRMSE) following ASHRAE Guideline 14 and the International Performance Measurement and Verification Protocol (IPMVP). Their results, obtained respectively with MinMaxScaler and StandardScaler preprocessing, are shown in Tables 4 and 5. It should be noted that all these error measures were calculated using electricity consumption values that had already been normalised to ensure model comparability. However, normalising the data does not affect the relative performance ranking of the evaluated methods. In both scenarios, the best predictive performance was achieved by the MLP model, with the lowest errors and a high R^2 score. Under Min-Max normalization, the MLP model achieved the highest predictive performance, with an RMSE of 0.0067, an MAE of 0.0052, and an R^2 of 99.61%. Moreover, the MLP model exhibits low error variability (CVRMSE = 3.47%) and negligible bias (NMBE = 0.70%), indicating robust and precise predictive performance. GBR and XGB follow, achieving comparable R^2 values of 99.27% and 99.21%, respectively. Nevertheless, both models incur

TABLE 3. Overview of the hyperparameter optimization process for each model.

Model	Hyperparameters	Search Space	Optimal Value
MLP	hidden_layer_sizes	(50,),(100,),(50,50), (100,100),(100,200)	(50,)
	alpha	0.1, 0.01	0.01
	activation	relu	relu
	solver	adam, sgd	adam
	max_iter	1000, 2000	1000
XGB	n_estimators	100, 200, 300	300
	max_depth	3, 5, 7	3
	learning_rate	0.01, 0.1, 0.3	0.3
GBR	n_estimators	50, 100, 200	300
	max_depth	2, 3, 4, 5	2
	learning_rate	0.1, 0.5	0.5
	loss	ls, lad, huber	huber
RF	max_depth	4-10	10
	n_estimators	10-50	10
DT	—	—	—
SVR	kernel	rbf, poly	poly
	C	0.1, 1.0, 10.0	1.0

higher error magnitudes than MLP, with RMSE values of 0.0093 for GBR and 0.0096 for XGBoost.

In contrast, the RF and DT models demonstrate reduced predictive capability under Min-Max normalization. RF attains an R^2 of 0.9534 with an RMSE of 0.0236, while DT records an R^2 of 91.26% and an RMSE of 0.0323. The SVR model exhibits the weakest performance in this configuration, with an RMSE of 0.0431 and an R^2 of 84.43%, suggesting limited suitability for this normalization approach. With standardScaler, its performance improved significantly for the best performing models. The MLP model demonstrates outstanding accuracy achieving an R^2 of 99.98%, an RMSE of 0.0114 and a CVRMSE of 0.72%, together with a negligible bias (NMBE = -0.17%). This strong performance can be attributed to the capability of neural networks to establish complex, non-linear relationships between the input features and the target variable.

GBR and XGBoost also perform well in terms of predictive accuracy when standardization is applied, with both models achieving R^2 values exceeding 0.99. Specifically, GBR attains an RMSE of 0.0759 and a CVRMSE of 4.76%, while XGBoost achieves an RMSE of 0.0792 and a CVRMSE of 4.97%. Despite their strong predictive capability, the dispersion of their errors is noticeably higher than that observed for the MLP model.

In contrast, SVR and RF exhibit a decline in performance under standardization. SVR records an RMSE of 0.1420 with an R^2 of 0.9747, whereas RF yields an RMSE of 0.1923 and an R^2 of 95.37%. The Decision Tree model demonstrates the weakest performance, with an RMSE of 0.2639 and an R^2 of 91.28%, highlighting its limited ability to capture complex nonlinear relationships. It is worth noting that, across both preprocessing strategies, all evaluated models exhibit

TABLE 4. Comparative analysis using MinMaxScaler.

Model	MAE	RMSE	R ²	NMBE (%)	CVRMSE (%)
MLP	0.0052	0.0067	0.9961	0.7004	3.4738
GBR	0.0060	0.0093	0.9927	-1.0374	4.7784
XGB	0.0066	0.0096	0.9921	-1.2166	4.9738
RF	0.0173	0.0236	0.9534	-6.8242	12.1097
DT	0.0214	0.0323	0.9126	-3.2190	16.5908
SVR	0.0371	0.0431	0.8443	9.5403	22.1448

TABLE 5. Comparative analysis using standardscaler.

Model	MAE	RMSE	R ²	NMBE (%)	CVRMSE (%)
MLP	0.0067	0.0114	0.9998	-0.1728	0.7177
GBR	0.0492	0.0759	0.9928	-1.0270	4.7640
XGBoost	0.0545	0.0792	0.9921	-1.2167	4.9739
SVR	0.0842	0.1420	0.9747	-2.0561	8.9199
RF	0.1414	0.1923	0.9537	-6.8373	12.0743
DT	0.1761	0.2639	0.9128	-2.8092	16.5713

normalized mean bias error (INMBE) values below 10% and coefficients of variation of RMSE (CVRMSE) below 30%, which are commonly adopted reference thresholds in ASHRAE Guideline 14 [61] and the International Performance Measurement and Verification Protocol (IPMVP) for hourly building energy model [62].

The results obtained reveal superior predictive accuracy compared to those reported in [60], which evaluated the same dataset with similar models, reporting an RMSE of 0.0165 and an R² of 94% for the MLP model. Admittedly, this difference in performance can be explained in particular by preprocessing strategies, hyperparameter tuning, and experimental configuration, but this comparison suggests that the chosen optimisation approach improved predictive performance.

It is important to note that integrating various explainable AI (XAI) techniques, including SHAP, LIME, and the Eli5 library to measure variable importance, did not directly affect predictive accuracy indicators (RMSE and MAE). In fact, these techniques were primarily intended to enhance the interpretability and transparency of the models by providing insight into the factors that influence their predictions. These quantitative indicators assess predictive performance without revealing the models' underlying decision-making mechanisms. Consequently, the following section focuses on explainability analysis to further explore and interpret how the model works.

B. EXPLAINABILITY ANALYSIS

The objectives of the two-pronged explainability analysis were: (i) to identify the characteristics that most influence electricity consumption forecasts, thereby enhancing predictive performance, and (ii) to improve the interpretability of the model by providing a transparent view of the decision-making process. To this end, we used two explainability methods, one global and the other local, providing a comprehensive and detailed understanding of the contributions of characteristics at both model and instance levels.

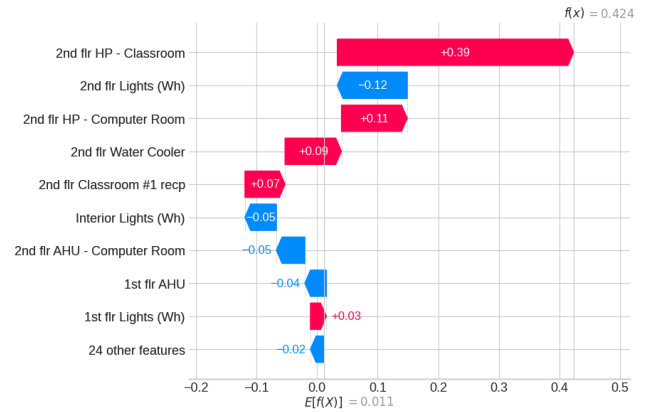


FIGURE 5. Local explanations provided by waterfall plot visual of SHAP for a specific sample data row 5. The red bar indicate that the corresponding feature increases the predicted electricity consumption for the analyzed instance, whereas the blue bars represent features that reduce the predicted value relative to the baseline expectation.

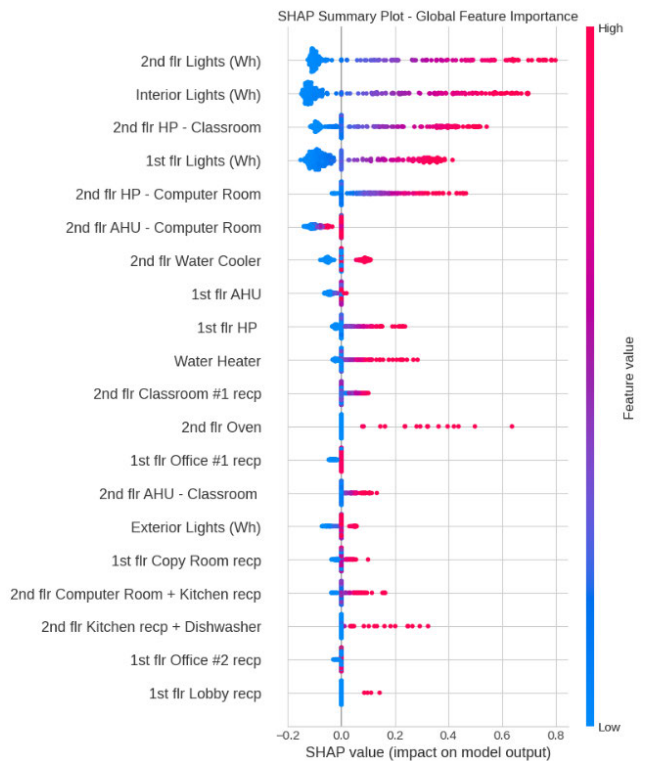


FIGURE 6. Global explanations provided by SHAP visual summary plot.

Global explainability seeks to highlight the characteristics that consistently influence the model's predictions on the entire dataset. This study used SHAP (SHapley Additive exPlanations) and the Eli5 library for global interpretation to identify the dominant contributors to electricity consumption across the entire dataset, while applying LIME and SHAP plots for local analysis. The complementary use of these explainability methods makes model interpretation more robust and reliable.

For global explainability, SHAP summary plots (Fig. 6) were used to rank features according to their overall contribution to predicting electricity consumption, based on the mean absolute Shapley values. In this representation, features are ordered in descending importance according to their average absolute SHAP impact on the model output. Positive SHAP values indicate features that increase the predicted electricity consumption, while negative values correspond to features that reduce the prediction. Each point represents an individual observation, and the color gradient (from blue to red) reflects the magnitude of the feature value.

The summary plot reveals that lighting and HVAC-related variables dominate the prediction structure. The most influential features include *2nd floor Lights (Wh)*, *Interior Lights (Wh)*, *2nd floor Heat Pump – Classroom*, *1st floor Lights (Wh)*, and *2nd floor Heat Pump – Computer Room*. For these variables, higher feature values (red points) are predominantly associated with positive SHAP values, indicating a consistent increase in predicted electricity consumption when their operational levels rise.

Furthermore, the dispersion of SHAP values is noticeably larger for these top-ranked variables, suggesting a stronger variability in their contribution across different operating conditions. In contrast, plug-load and specific receptacle circuits exhibit SHAP values more concentrated around zero, indicating comparatively lower global influence. Overall, the SHAP analysis confirms that HVAC systems and lighting subsystems are the primary drivers of total building electricity consumption within the predictive model.

The results obtained from the Eli5 feature importance analysis (Figures 7 and 8) are consistent with the SHAP findings, emphasizing that thermal conditioning systems (HVAC) and lighting subsystems serve as the principal drivers of total electricity consumption in the building. In contrast, plug-load circuits and individual receptacle branches exhibit markedly lower importance scores, suggesting a more localized and context-specific influence. The consistency across these global explainability methods confirms the reliability of the identified influential features. In a practical approach, the global explainability analysis revealed that HVAC lighting components reflects their structural significance in overall building energy demand, as they constitute the primary end-use categories in commercial facilities. This increased consumption on the second floor is attributable to higher occupancy rates and extended opening hours, which generally run from 8:30 a.m. to 5:00 p.m. These observations make it possible to identify targeted energy management measures, including optimizing HVAC use according to needs and reducing lighting consumption in heavily frequented areas.

In order to assess the potential influence of correlated predictors on feature attribution, we calculated the conditional importance of permutation and compared it to the standard importance of permutation Fig. 9). The conditional approach reduces the bias caused by

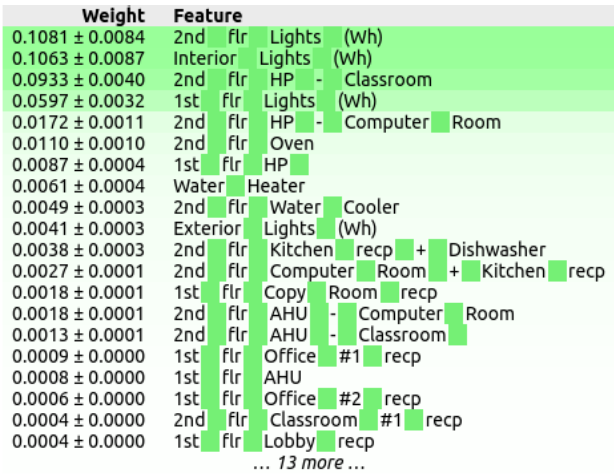


FIGURE 7. Global explanations provided by Eli5 using MLP model.

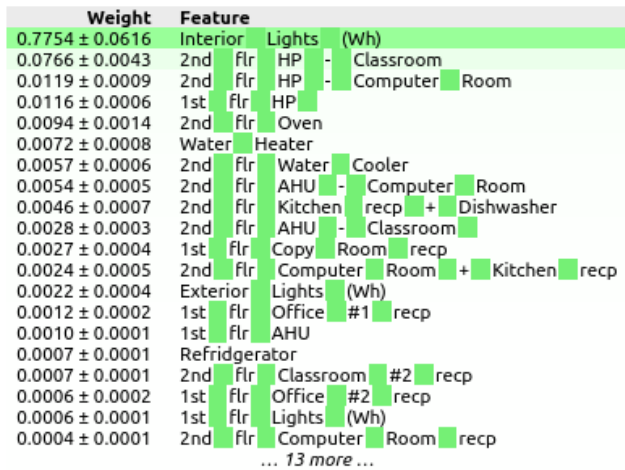


FIGURE 8. Global explanations provided by Eli5 using XGB model.

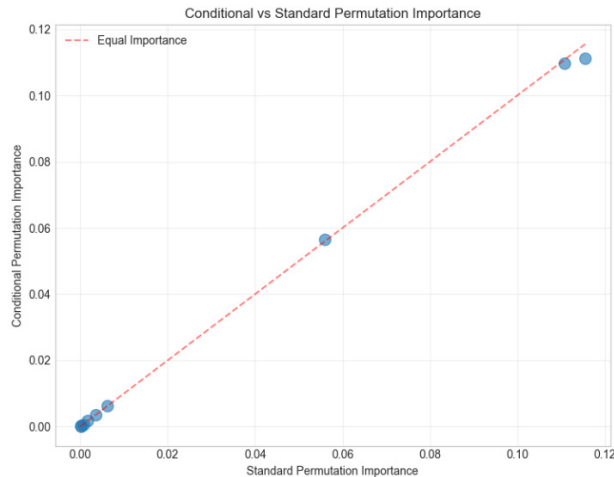


FIGURE 9. Comparison between standard and conditional permutation importance, illustrating the stability of feature rankings in the presence of correlated predictors.

multicollinearity by keeping the dependency structure between correlated features during permutation. As shown

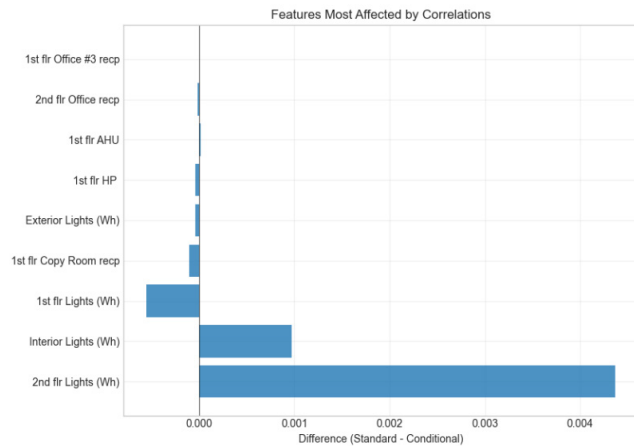


FIGURE 10. Difference between standard and conditional permutation importance (Standard – Conditional) for the most affected features, highlighting the limited impact of feature dependencies on attribution magnitude.

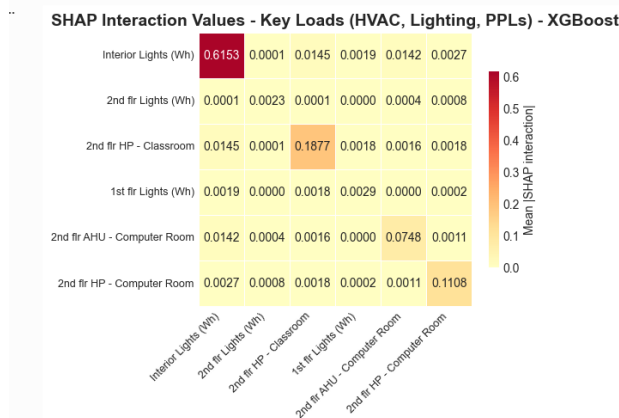


FIGURE 11. Heatmap of SHAP interaction values. Each cell represents the mean absolute SHAP interaction value for a pair of features. Diagonal elements correspond to the main (individual) effects of each feature, whereas off-diagonal elements quantify pairwise interaction effects between features.

in Fig. 10, the feature rankings are very consistent between the two methods, while the differences in magnitude are minimal. Importantly, the dominant loads related to HVAC and lighting components maintain their relative importance despite the dependencies between features. These findings are aligned with best practices for interpreting machine learning models under correlated inputs and confirm that the influential predictors identified are not artifacts of collinearity effects.

For a more detailed examination of whether interactions between characteristics have any effect, SHAP interaction values quantify the magnitude of pairwise contributions between input features and model output. As illustrated in Fig. 11, the majority of off-diagonal elements have relatively small values, indicating that model predictions depend primarily on the individual effects of features rather than strong interaction terms. However, some interactions

between HVAC and lighting variables show moderate interaction strengths, which suggests that dependency patterns are localized within building subsystems. It is important to note that these interaction amplitudes remain significantly lower than the corresponding main effects (diagonal elements), confirming that the overall attribution results are not primarily driven by interaction biases induced by multicollinearity.

Local explainability was conducted to interpret individual predictions on an hourly basis. To this end, we applied LIME and SHAP waterfall plot to a representative data sample (line 5). The LIME method provides more accurate explanations by estimating the model's behavior around a particular case, thereby highlighting each feature's contribution to the forecast. As illustrated in Fig. 12, orange indicates positive features contributions to electricity consumption, while the color blue corresponds to a negative contribution. Similarly, the SHAP waterfall plot (Fig. 5) provides an

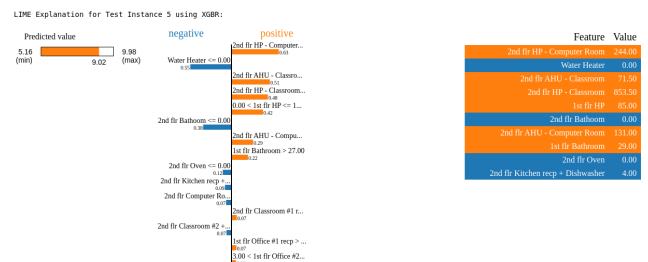


FIGURE 12. Local explanations provided by LIME using XGBR to explain instance row=5 in the testing set.

additive decomposition of the model output, starting from the expected prediction $E[f(X)]$ and sequentially incorporating feature-specific Shapley values to reach the final predicted value $f(x)$. In this representation, red bars indicate positive SHAP contributions that increase the predicted electricity consumption, whereas blue bars represent negative contributions that decrease the prediction relative to the baseline.

Both LIME and SHAP consistently identify *2nd floor AHU – Computer Room* and *2nd floor Heat Pump – Classroom* as among the most significant contributors to increased electricity consumption in the analyzed instance, particularly for the GBR and MLP models. The waterfall plot further quantifies the magnitude of these contributions, showing how dominant HVAC components shift the prediction upward, while other subsystems partially offset this increase.

Local explanations demonstrate that the significance of features varies across different observations, highlighting the importance of interpretability at the instance level for analysing trends in electricity consumption. The relative influence of HVAC, lighting, and plug-load circuits changes depending on the operating condition represented by each instance.

From an energy perspective, this finding is logically consistent. HVAC components, especially heat pumps and air handling units, generally represent high-power systems that

are responsible for thermal regulation, and their operation significantly impacts the building's total electricity demand. When operating at high load levels, their contribution to overall consumption becomes predominant. In contrast, lighting and auxiliary loads can have a positive or negative impact depending on their instantaneous operating condition relative to the baseline.

Local explanations further indicate that the contributions of characteristics are specific to each observation, reflecting the inherently context-dependent dynamics of electricity consumption. The relative influence of HVAC, lighting, and outlet load circuits varies depending on the particular operating state represented by each instance. This variability at the instance level highlights the importance of local interpretability for capturing short-term fluctuations in building energy consumption.

A quantitative assessment of the consistency between local and global explainability approaches, as well as their alignment with exploratory data analysis, is presented separately in the consistency section.

Table 6 shows the state-of-the-art recent studies on electricity consumption prediction using machine learning, and more specifically, how explainability techniques are addressed. For the most part, existing studies employ a limited number of predictive models and rely primarily on a single explainability method, typically SHAP or feature importance, to evaluate feature contributions. While these approaches provide insight into model behavior, they are often limited to global or local explainability.

In contrast, this study adopts a multi-model, multi-explainability framework that integrates multiple regression models, including GBR, XGB, MLP, SVR, random forest (RF), and DT. Numerous additional explainability techniques, including LIME, SHAP, and Eli5, are jointly used to analyse electricity consumption estimates. This concept supports both local and global interpretability, enabling a more comprehensive analysis of feature contributions across different models.

Compared to previous studies, including those of [34], [38], [42], and [54], which primarily focus on SHAP-based explanations, the proposed approach leverages the complementary strengths of multiple XAI methods. Specifically, SHAP provides both global and local feature explanations, LIME delivers instance-level interpretability, and ELI5 supports model-level feature importance analysis. The combination of these techniques mitigates the limitations associated with reliance on a single explainability method.

Furthermore, in contrast to works such as [37], which mainly aim to identify significant features, the proposed methodology extends the analysis by evaluating feature contributions across multiple predictive models and explainability techniques. This comprehensive assessment of interpretability is particularly relevant for electricity consumption forecasting applications, where transparent and well-justified

predictions are essential for informed decision-making and stakeholder trust.

C. QUANTITATIVE CONSISTENCY BETWEEN XAI AND EDA

To assess the level of consistency between SHAP-based global feature rankings and correlation-based rankings obtained through exploratory data analysis (EDA), Spearman's rank correlation coefficient (ρ) was employed. This non-parametric coefficient evaluates the monotonic association between two ranked variables. Additionally, the Overlap@k metric was used to quantify the proportion of shared features among the top-k ranked variables identified by SHAP and EDA. These metrics were calculated for each predictive model and further examined across individual load categories (HVAC, Lighting, and Plug and Process Loads—PPLs). Table 7 summarizes the agreement between SHAP-based global rankings and correlation-based EDA rankings using Spearman's rank correlation (ρ).

HVAC features show moderate positive correlations, with the strongest alignment for MLP ($\rho = 0.643$) and XGBoost ($\rho = 0.619$), indicating consistent prioritization across methods. Lighting exhibits the highest agreement, with perfect rank consistency for MLP and SVR ($\rho = 1.000$), reflecting stable and robust contributions.

In contrast, Plug and Process Loads (PPLs) display weaker correlations ($\rho = 0.273$ – 0.430), likely due to their larger feature set and lower, more variable contributions.

Overall, SHAP-derived feature importance is largely consistent by correlation-based EDA, particularly for dominant load categories, reinforcing the robustness of the proposed explainability framework.

1) QUANTITATIVE CONSISTENCY BETWEEN XAI TECHNIQUES

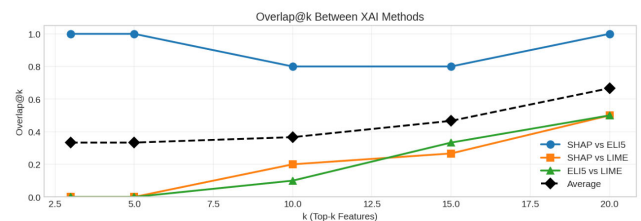


FIGURE 13. Consistency metric for various XAI techniques.

To evaluate the alignment between different explainability techniques, the Overlap@k metric was calculated between SHAP, ELI5 (permutation importance) and LIME. This metric quantifies the proportion of common features among the variables ranked among the top k identified by each pair of XAI methods. Fig. 13 shows the Overlap@k analysis between XAI methods over increasing values of k. The highest consistency is observed between SHAP and ELI5, with complete overlap for the top 3 and 5 features and

TABLE 6. Comparative analysis of ML and XAI methods in previous studies.

Study	ML Models	Explainable Approach	LIME	SHAP	ELI5	Methodology / Impact of XAI
[38]	RFR	—	X	X	—	PV power forecasting; improved model quality
[40]	XGB	—	—	—	X	Predict solar PV generation; improved model quality
[49]	—	—	—	X	—	Predict energy consumption; identify influencing factors
[50]	LSTM-XGB	—	—	X	—	Energy consumption of electric bus; influence of features
[34]	XGB	—	—	X	—	Forecast energy use; contributions of features
[60]	SVR, LR, RFR, MLPR, GBR	—	X	X	X	Electricity consumption prediction; feature contribution
[42]	RF	—	—	X	—	Predict energy use; influence of features
[54]	XGB	—	—	X	—	Understand impact of input features
[37]	LSBoost, SVR	—	—	X	—	Detect relevant features; improve model quality
[35]	XCNN, LSTM	—	—	X	—	Forecast energy consumption; influence of features
Our study	GBR, MLP, XGB, SVR, RF, DT	Data understandability				
		Data visualization				
		Data transformation	X	X	X	Electricity consumption forecasting; model interpretability
		Local interpretability				
		Global interpretability				

TABLE 7. Spearman rank correlation (ρ) and associated p -values between SHAP global feature rankings and EDA-based rankings per load category and predictive model.

Category	MLP		RF		SVR		GBR		XGBoost	
	Spearman ρ	p	Spearman ρ	p	Spearman ρ	p	Spearman ρ	p	Spearman ρ	p
HVAC	0.643	0.0856	0.571	0.1390	0.524	0.1827	0.381	0.3518	0.619	0.1017
Lighting	1.000	0.0000	0.200	0.8000	1.000	0.0000	0.400	0.6000	0.400	0.6000
PPLs	0.313	0.1671	0.409	0.0656	0.273	0.2317	0.430	0.0518	0.384	0.0853

consistently strong alignment for higher k values. The agreement between SHAP and LIME, as well as between ELI5 and LIME, gradually increases as k increases, which indicates convergence in identifying wider sets of informative features. Overall, the results demonstrate stable alignment between methods, particularly for the most influential features.

D. GENERALIZABILITY ACROSS CLIMATIC CONTEXTS

Even though this study focuses on a building in Virginia, which has a temperate climate, our methodology isn't dependent on climate at the structural level. Predictive architecture and explainability methods work based on electrical measurements at the circuit level rather than climate-specific parameters.

However, the energy consumption of buildings, particularly heating, ventilation, and air conditioning loads, is inherently dependent on climate. In colder climates (e.g., Minnesota), the dominance of heating would likely push the SHAP ranking toward heating-related variables, while in warmer climates (e.g., Florida), cooling subsystems would have higher weights.

Consequently, while the explainable predictive framework is transferable, the extent and order of influential features may vary depending on the climate zone. Validation

across multiple climates is a necessary step to strengthen generalization.

V. CONCLUSION

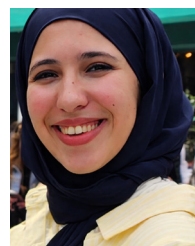
The main objective of this study was to demonstrate the value of integrating interpretability throughout the machine learning process for electricity consumption forecasting. The proposed methodology follows a structured three-step process: data understanding, predictive modeling, and systematic interpretation. It ensures that forecasting accuracy is supported by transparent and physically consistent explanations, which enhances the reliability of data-driven decision-making in smart buildings. To forecast electricity consumption, several machine learning models were implemented and evaluated comparatively. Among the evaluated models, MLP achieves the best predictive performance, while XGBoost and GBR also demonstrated strong results. Beyond accuracy, explainability techniques, including SHAP, LIME, and ELI5, were applied to identify key energy drivers. A quantitative consistency analysis using Spearman's rank correlation and Overlap@ k measures demonstrated moderate to strong agreement between SHAP-based rankings and correlation-based exploratory data analysis, particularly for HVAC and lighting subsystems. These results confirm that the identified influential features are not dependent on the method.

Furthermore, the proposed framework not only achieves high statistical predictive performance; it also complies with internationally recognized calibration standards (ASHRAE Guideline 14 and IPMVP), which ensure methodological robustness and operational credibility for building energy performance evaluation. The experimental results confirm the proposed research hypotheses; the H1 is validated as explainable ML models (MLP and XGBoost) achieved competitive predictive performance while satisfying ASHRAE Guideline 14 thresholds. In addition, SHAP, LIME, Eli5 consistently identified HVAC components and Lighting as discriminative contributors to electricity consumption (H2). Furthermore, the quantitative consistency analysis demonstrates stable interpretability results that enhance transparency (H3). The proposed explainable framework is generic and transferable due to its model-independent and data-driven configuration, making it applicable to other smart buildings and energy consumption-related datasets beyond our case study. The current research shows that an explainable approach can be used to validate the chosen predictive model and enhance the relevance and reliability of its predictions. The gap between complex data-driven insights and clear, transparent, and understandable interpretations will be bridged by adopting an explainable approach for building AI models. Despite these contributions, the present study has certain limitations that need to be openly acknowledged. The analysis is primarily conducted on data collected from a single commercial building to allow a detailed explainability analysis of internal energy drivers. Another limitation, is the non integration of explicit climatic and occupancy variables, enabling the study to emphasize the interpretability of consumption patterns derived from building-level measurements. In future research, we are convinced to extend the proposed explainable framework to multiple buildings with different usage profiles and climatic conditions. The integration of weather and occupancy variables will be addressed to enhance the robustness and generalizability. Furthermore, transforming SHAP and LIME visualizations into more interpretable, user-friendly formats will enable deeper insights and support practical energy management.

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